CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 8: Predicting What Comes Next? Exploring Sequences and Progressions

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write the first four terms of the sequence $t_n = n^2 + 1$.
2. For $t_n = 7n - 4$, find the 15th term. Is 150 a term?
3. Write the first four terms of $s_1 = 4$, $s_n = 2s_{n-1} - 3$.
4. Find the nth term and the 30th term of the AP $9, 14, 19, \ldots$
5. Which term of the AP $50, 46, 42, \ldots$ is -10?
6. The 5th term of an AP is 22 and the 12th term is 50. Find a and d.
7. Find the sum of the first 60 natural numbers.
8. Find $21 + 22 + \ldots + 50$.
9. Find the 8th term of the GP $3, 6, 12, \ldots$
10. Which term of the GP $4, 12, 36, \ldots$ is 2916?

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Answer Key

1. 2, 5, 10, 17
2. $t_{15} = 101$; yes, 150 is the 22nd term
3. 4, 5, 7, 11
4. $5n + 4$; 154
5. The 16th term
6. $d = 4$, $a = 6$
7. $\frac{60 \times 61}{2} = 1830$
8. $1275 - 210 = 1065$
9. $3 \times 2^7 = 384$
10. $3^{n-1} = 729 = 3^6$, so it is the 7th term

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